

and Colorado Springs, Colorado, along with monitoring stations operated by the National Geospatial-Intelligence Agency (NGA). The master control station is located at Schriever Air Force Base in Colorado Springs. The tracking information from the monitoring stations is sent to the master control station. The master control station is operated by the 2nd Space Operations Squadron (2 SOPS) of the United States Air Force (USAF). The 2 SOPS use the ground antennas located at Ascension Island, Diego Garcia, Kwajalein, and Colorado Springs to provide a regular navigational update to each GPS satellite. These updates synchronise the atomic clocks on board the satellites to within a few nanoseconds of each other. The inputs from the ground monitoring stations, space weather information, and various other inputs are used by a Kalman Filter to provide the updates.

The user's GPS receiver is the user segment (US) of the GPS. In general, GPS receivers are composed of an antenna, tuned to the frequencies transmitted by the satellites, a clock that is sufficiently accurate so that there is very little error, and a processor that interprets the incoming signals. Some GPS receivers also include a display for providing location and speed information to the user. The number of channels, i.e. the

Trilateration is the technique used by GPS receivers to pinpoint the exact location on Earth

number of satellites it can monitor simultaneously, is often used to describe a GPS receiver. This is a figure that is gradually increasing. Earlier, the number of channels was limited to four or five, but

nowadays it is standard for a GPS receiver to have between 12 and 20 channels.

In order to pinpoint the exact location anywhere on Earth, the GPS receivers use a technique known as trilateration. In case the receiver receives signals from three satellites, the exact coordinates on the Earth's surface can be provided. However, if there are four satellites, then, the altitude can also be measured. The basic principle of working of trilateration is as follows.

All GPS satellites are time synchronised with each other with the help of caesium atomic clocks. Thus the maximum time error between the satellites is a few nanoseconds. These GPS